University of Florida, Department of Mathematics

Algebra PhD Exam, September 1999

Please do 7 out of the 11 problems below.

1. Here $\mathbb{Z}_n$ denotes the cyclic group of order n.

(a). Let p be a prime. How many subgroups of order p^2 does the abelian group $\mathbb{Z}_{p^3} \oplus \mathbb{Z}_{p^2}$ have?

(b). What are the elementary divisors of the group $\mathbb{Z}_{26} \oplus \mathbb{Z}_{42} \oplus \mathbb{Z}_{49} \oplus \mathbb{Z}_{200}$? What are its invariant factors?

(c). Determine up to isomorphism all abelian groups of order 20.

2. This problem has two parts.

(a). Exhibit an automorphism of the cyclic group of order 6 that is not an inner automorphism.

(b). Show that if a normal subgroup N of order p (p prime) is contained in a group G of order p^n, then N is in the center of G.

3. This problem has three parts.

(a). Is the additive group $\mathbb{Q}$ of rationals a divisible abelian group?

(b). Show that no nonzero finite abelian group is divisible.

(c). Show that no nonzero free abelian group is divisible.

4. Let R and S be rings and $A_R, {}_RB_S, C_S$ (bi)-modules. Demonstrate an isomorphism of abelian groups between $\mathrm{Hom}_S(A \otimes_R B, C)$ and $\mathrm{Hom}_R(A, \mathrm{Hom}_S(B, C))$. (You should state very clearly what must be checked.)

5. Let R be a ring, let

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

be a short exact sequence of left R-modules and let D be a right R-module. Show that

$$0 \longrightarrow D \otimes_R A \xrightarrow{1_D \otimes f} D \otimes_R B \xrightarrow{1_D \otimes g} D \otimes_R C \longrightarrow 0$$

is a short exact sequence of abelian groups under each one of the following hypotheses:

(a). $0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$ is split exact.

(b). R has an identity and D is a free right R-module.

(c). R has an identity and D is a projective unitary right R-module.

6. Let A be a commutative ring with 1 such that every element $x \in A$ satisfies $x^n = x$ for some $n > 1$, depending on x. Prove that

(a). every prime ideal of A is maximal.

(b). the Jacobson radical of A is (0).

7. Let C and D be categories, and let S and T be covariant functors from C to D.

(a). Define a natural isomorphism $\alpha : S \to T$.

(b). If B is a unitary left module over a ring R with identity, show that there is a "natural" isomorphism of modules

$$\alpha_B : R \otimes_R B \cong B.$$

8. This problem has two parts.

(a). Give an example of commutative rings R, A with $R \subset A$, where A is Noetherian but R is not.

(b). Let $R = K[x, y]$ be the polynomial ring in two variables over the field K, let A denote the ideal (xy, y^2) in R and let $c \in K$. Show that

$$A = (y) \cap (x + cy, y^2)$$

and that this is a primary decomposition.

9. This problem has two parts.

(a). Show that if S is a multiplicative subset of a Dedekind domain R (with $1 \in S$, $0 \notin S$), then $S^{-1}R$ is a Dedekind domain.

(b). Show that if I is a nonzero ideal in a Dedekind domain, then R/I is an Artinian ring.

10. Let p be a prime and $n \geq 1$ an integer. Show that F is a finite field with p^n elements if and only if F is a splitting field of $(x^{p^n} - x)$ over the field $\mathbb{Z}_p$ of p elements.

11. This problem has two parts.

(a). Give an example of fields $M \supset L \supset K$ such that M/L and L/K are normal extensions, but M/K is not normal. Justify your answer.

(b). Let $L \supset K$ be fields such that $[L : K] = 8$. Prove that there exist $\alpha, \beta, \gamma \in L$ such that $L = K(\alpha, \beta, \gamma)$.